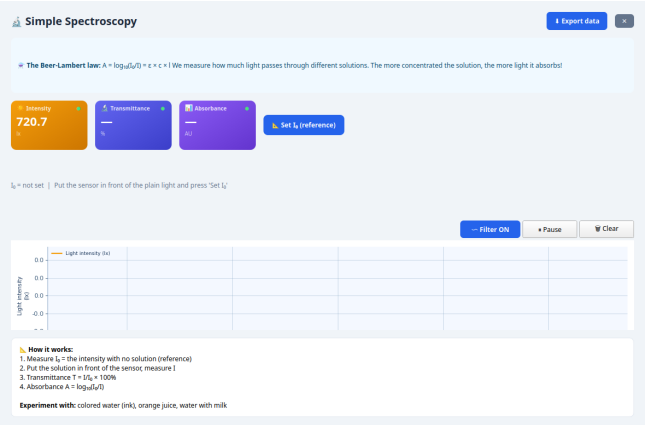
Spectroscopy

Transmittance, absorbance and the Beer–Lambert law — with a sensor that cannot tell colours apart

Level: Scientist — ages 15–19 (Romanian school years IX–XII) demonstration, 50 min investigation, or a 2–3 week project

- **Time in class:** 20 min
- **Sensors:** The light sensor (illuminance)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Scientist — ages 15–19 (Romanian school years IX–XII), but with a **precise boundary inside that range**: absorbance is a logarithm, and **logarithms enter the Romanian mathematics syllabus only in year X**. In year IX the activity runs whole and honestly *on transmittance alone*, in per cent, which is a plain ratio; from year X absorbance and the Beer–Lambert law join it. The most direct entries are in **year IX** (chemistry: solutions and concentration) and **year X** (mathematics: logarithms); the Beer–Lambert law and colorimetry are **not in the core syllabus** but enrichment, science club and olympiad content.

Assumed knowledge

Preparing a dilution and the idea of concentration. For absorbance: **the base-ten logarithm**, so from year X. For the calibration curve: an XY chart and a line of best fit in a spreadsheet.

Time in class

Short: 20 min (demonstration). Full: 50 min (a whole lesson, with the filters and the dilution series). Project: 2–3 weeks, about four lessons plus repeats.

Teacher preparation

40 min — **build and test a rig yourself**, because the geometry is everything: lamp, sample and sensor on a fixed line, at fixed distances, inside something that excludes room light. A shoebox on its side, with a hole for the lamp at one end, a slot in the middle for the cuvette and the sensor at the other end, is entirely adequate and takes ten minutes. Mark the positions with tape. Prepare and label the stock solutions. And close the blinds.

Sensor settling time

5 seconds after each change of sample — the light sensor responds practically instantly. **The real wait is the lamp's**: a filament lamp needs several minutes to settle, and a torch on batteries fades continuously. Switch the lamp on ten minutes before the lesson.

Sensors and boxes The light sensor in the box. **It is a broadband sensor, not a spectrometer**: it gives one number, the illuminance, and cannot say *which* wavelengths were absorbed. One box per group of 3–4 pupils.

Three routes

Short 20 min: set the reference with a clear path, insert coloured filters one at a time and read the transmittance, then a strongly coloured solution and a diluted one, and discuss why absorbance is the useful quantity.

Full 50 min: rig and reference (8), investigation 1 with filters (12), investigation 2 with the dilution series (20), export and plot the calibration line (10).

Project 2–3 weeks: a repeatable rig and its characterisation, the filters, the full calibration with an unknown determined, then the calibration repeated on another day and with another lamp, for reproducibility.

Read this before you plan the lesson

The light sensor is **broadband**: it reports one number, the illuminance, weighted roughly to the response of the human eye, and it **cannot say which wavelengths were absorbed**. "Spectroscopy" here therefore means **investigating transmission and absorption with coloured filters and coloured solutions**, not obtaining a spectrum.
This is a genuine limitation and it must be told to pupils **in the first minute**. Handled well it becomes the most instructive part of the activity rather than a disappointment: Section 5 shows how to work around it honestly, by putting a coloured filter in the light path *before* the light reaches the sensor — which is exactly how a *filter photometer* works, a real instrument used for decades, and exactly what a school colorimeter is.

1. What the app shows and what it does not

Unlike the guided activities, this is not a step-by-step wizard. It is an **open instrument**: it turns a light reading into transmittance and absorbance against a reference you set yourself, and leaves the chemistry to you. There is no green button, no **Back**, no **Finish**. **You set the pace of the lesson**, and the investigations in section 6 are ready to run exactly as written.

There are three cards on screen and one button that changes everything:

- **Intensity I**, in `lx` — the only *measured* quantity.
- **Transmittance**, in per cent, with one decimal — calculated, $T = 100\, I/I_0$.
- **Absorbance**, dimensionless, with three decimals — calculated, $A = \log_{10}(I_0/I)$.
- **The Set I_0 (reference) button**, pressed with the blank in the beam. For solutions the blank is **the same cup filled with clear water** — which is what the app calls it too; for filters the blank is the clear path, with no filter. Without a reference, the other two cards have nothing to show.

Underneath sits the **Light intensity** chart, and on the right a fixed method panel: **How to measure** (the blank, the sample in the same place, back to the blank every two or three samples, a darkened room), the two formulas with every symbol explained, and **What to measure** — clear solutions diluted step by step from one stock, preferably a red or magenta dye with a white lamp, or a coloured LED of the colour the dye absorbs. The panel also warns that milk and orange juice scatter light instead of absorbing it, and that the absorbance should be kept under about 1. Pupils can read it on their own; this guide expands on it.

The skill that transfers out of this activity is **referencing a measurement against a blank**. Absorbance means nothing in isolation; it exists only *relative to a reference*. Pupils who genuinely internalise this understand a large part of analytical science.

Technical note — the reference, the two calculated cards, and what is not exported

The sensor. The light sensor in the box, a broadband one (the `LUX` column in the exported files). With firmware 2.2.5 or newer, the box chooses its own sensitivity and measures from a few thousandths of a lux up to direct sun (about 120 000 lx). **Before 2.2.5 the reading stopped at about** 3800 **lx** — and a desk lamp close to the sensor easily exceeds that, in which case I_0 is false and every absorbance comes out too small. With old firmware, move the lamp back until I_0 stays below about 3000 lx, or reflash the box. The app helps: if it knows the box's firmware is old and you set a reference above 3000 lx, it shows a yellow strip asking for exactly that.

Setting the reference. With the blank in the beam, press **Set I_0 (reference)**. Until then, the line under the cards says *I_0 = not set — Put the cup with clear water (the blank) in the light and press "Set I_0"*; after the press it shows the stored value, $I_0 = \ldots$ lx, and reminds the pupil to write it down, because the exported file holds only the lux column. **There is no clear button**: you reset it by pressing again. **The reference does not survive between sessions** — close the app and it is gone.

If pressing it does nothing, the reading at that moment was zero or missing: the app stores the reference only if the illuminance is strictly positive, and the line stays on "not set". Switch the lamp on and try again.

More light than the reference. If the sample lets through more light than the reference ($I > I_0$) — because the lamp has warmed and brightened, because something has moved, or because room light has crept in — the absorbance would come out negative and the transmittance above 100%. The app does not show such values: the **Transmittance** and **Absorbance** cards stay blank, and a yellow strip says *$I > I_0$: more light passes than through the blank. Set the reference again.* (Up to 3% above I_0 counts as sensor noise and shows as 100% and 0.) **This is not a fault: it is the signal that the reference is no longer valid.** In the same kind of strip the app says *No light reaches the sensor: check the lamp and the cup* when the reading is zero. Teach the class to recognise the signal and reset the reference, not to report the value.

The chart. A single chart, **Light intensity**, in lux, with **Smooth: on/off**, **Pause** and **Clear**. It shows only the last two minutes, with the time axis in minutes:seconds. When several boxes are connected, the **Change box** list in the header picks the box being read; switching box drops the reference.

The export — and what is missing from it. The **Export data** button, top right, writes an `.xlsx` (the default) or `.csv` file whose suggested name already carries the box and the date. **The file contains the lux column, not the transmittance and not the absorbance**: those are computed in the app, on screen, and saved nowhere. *This is not a gap but an opportunity*: pupils recompute T and A themselves from the raw column in a spreadsheet, which is a better exercise than reading the card — **but they must write I_0 down on paper**, because otherwise they have nothing to recompute them from. Independently of the button, the app also writes a raw CSV log of the stream to the `logs` folder; it does not delete it by itself unless a number of days has been set under **Keep logs (0 = always)** in **Settings** — and even then it asks at start-up before deleting.

2. Where this fits in the curriculum

- **Chemistry** — solutions and concentration (percentage concentration in lower secondary, molar concentration in **year IX**): the dilution series is the practical from the syllabus, but with an instrument that says whether the dilution came out right. Colorimetry and the determination of an unknown concentration from a calibration curve — the real analytical method, done with the class's own apparatus — are **not in the core syllabus**: they are enrichment content, suited to a chemistry club, olympiad preparation and specialist classes. Check the syllabus in force for your year and stream.

- **Mathematics — the base-ten logarithm, from year X**, with a physical meaning the textbook cannot give: $A = 1$ means a tenth of the light gets through, $A = 2$ a hundredth. Linear regression on the calibration series, interpolation for the unknown, and the limits of proportionality also appear.
- **Physics** — light intensity, absorption and scattering, transmission through a medium, the inverse-square law (investigation 4) and the spectral response of a detector; in upper secondary, again as enrichment rather than a compulsory syllabus topic.
- **Biology** — pigment absorption, chlorophyll and why leaves are green, turbidity as a proxy for cell density.
- **Cross-curricular** — instrument calibration, the idea of a *blank*, and the difference between a measured quantity and a derived one.

3. Time plan

Route	Time	What happens
Demonstration	**20 min**	Set the reference with a clear path (3). Insert coloured filters one at a time and read the transmittance for each (8). Insert a strongly coloured solution and then a diluted one (5). Discuss why absorbance, and not transmittance, is the useful quantity (4).
Full	**50 min**	Rig and reference (8), investigation 1 with filters (12), investigation 2 with the dilution series (20), export and plot the calibration line (10). The rig is built beforehand by the teacher. One box per group.
Project	**2–3 weeks**	About four lessons plus repeats; the schedule is below.

3.1 Schedule for the multi-week project

When	Time	What happens
Week 1, lesson 1	50 min	Build a repeatable rig: fixed lamp, fixed cuvette position, room light excluded. Characterise its stability and its noise.
Week 1, lesson 2	50 min	The filter survey and the complementary-colour investigation.
Week 2, lesson 3	50 min	The full dilution series, the calibration curve and the determination of an unknown concentration. Export.
Week 2, homework	20 min	Recompute T and A from the raw lux column in a spreadsheet and check them against what the app displayed.
Week 3, lesson 4	50 min	Repeat the calibration on another day with a fresh reference and compare with week 2. Discuss instrument drift and reproducibility. Present the findings.

4. Materials

Item	Per group	Notes
MagicBox with the light sensor	1	Note which face of the box the sensor's window is on; it must face the lamp squarely. The box is not waterproof: keep it outside the pouring area.
Laptop with the app	1	USB or WiFi.
Coloured filters or lighting gels	5–8	Red, orange, yellow, green, cyan, blue, magenta, plus a neutral grey if you have one. Theatre gel swatch books are cheap and ideal; coloured acetate or sweet wrappers work at a push.
Coloured solutions	3–5	Food colouring, ink, very dilute potassium permanganate, copper sulfate, tea or blackcurrant squash. Choose strongly coloured, *transparent* solutions — not cloudy ones. **With white light, a red or magenta dye gives the biggest signal**: it absorbs green, where the sensor is most sensitive. A blue or green dye absorbs in the red and gives a small signal — that is where a coloured filter helps. Milk and orange juice are no good for Beer–Lambert: they scatter light rather than absorb it.
Identical transparent containers	6+	Plastic cuvettes, sample tubes or straight-sided jars. **Identical is essential**: the container is part of the light path.
A steady lamp	1	A mains-powered desk lamp is best, because it does not dim as batteries drain. If using a torch, fit fresh batteries and do not change them mid-experiment.
Dark box or black cloth	1	To exclude room light. **This is the single biggest improvement you can make to data quality.** A shoebox with two holes works perfectly.
Ruler	1	To fix and record the lamp-to-sensor distance and to measure the path length through the container.
Measuring cylinders or syringes	2	For accurate dilution. A 10 mL syringe gives far better dilutions than a beaker and a guess.
Distilled or tap water	1 L	The solvent, and the blank.
Labels and marker	1 set	A dilution series without labels becomes uninterpretable within minutes.
Gloves and eye protection	1 set each	For handling dyes and permanganate.

Preparation. Build and test one rig *before* the lesson. The geometry is everything: lamp, sample and sensor must sit on a fixed line, at fixed distances, inside something that excludes room light. Mark the positions with tape so that nothing moves. Check that the reference reading is comfortably within the sensor's range and reasonably steady — **if it drifts by more than a few per cent over a minute, find the cause before the lesson rather than during it**. Prepare and label the stock solutions; the dilution series goes much faster if pupils dilute from a known stock than if they mix from scratch. And close the blinds: **a passing cloud changes the room light by far more than a dilute solution changes the transmission**.

5. Scientific background for the teacher

Transmittance, absorbance and Beer–Lambert

When light of intensity I_0 passes through an absorbing medium, the intensity that emerges is I. Two quantities describe this:

$$T = \frac{I}{I_0} \quad \text{(usually quoted as a percentage)}, \qquad A = \log_{10}\left(\frac{I_0}{I}\right) = -\log_{10} T.$$

Absorbance is the useful one, because it is the quantity that behaves *linearly*. The Beer–Lambert law states that

$$A = \varepsilon\, c\, l,$$

where ε is the molar absorption coefficient of the absorbing species (constant for a given substance at a given wavelength), c the concentration and l the path length through the sample. A plot of A against c is a straight line through the origin, so an unknown concentration can be read off a calibration line — exactly what a school colorimeter does.

The law follows from the fact that each thin slice of solution removes the same *fraction* of the light reaching it, so the intensity falls exponentially with distance, $I = I_0 \mathrm{e}^{-\alpha l}$, and the logarithm linearises it. Useful landmarks: $A = 0$ means all the light passes ($T = 100\%$); $A = 1$ means 90% is absorbed ($T = 10\%$); $A = 2$ means 99% is absorbed ($T = 1\%$).

Beer–Lambert needs light of a single wavelength. With white light and a broadband sensor, **the main cause of deviation is polychromatic light**: the wavelengths the dye absorbs strongly disappear first, and at high concentration what reaches the sensor is mostly light the dye barely absorbs, so the absorbance grows ever more slowly. The law also fails at very high concentration, where molecules interact and ε ceases to be constant, and whenever stray light reaches the detector without passing through the sample. All three bend the calibration line downwards at the top — the classic curved colorimeter plot.

5.1 Complementary colours: what a coloured solution is actually doing

A solution looks coloured because it *removes* the complementary colour from white light. Copper sulfate looks blue because it absorbs in the orange and red; potassium permanganate looks purple because it absorbs strongly in the green. **The colour you see is what is left over.**

The solution looks	It absorbs mainly	So the most sensitive filter is
Red	Cyan / blue-green	Cyan or green
Orange	Blue	Blue
Yellow	Blue / violet	Blue
Green	Magenta / red	Red
Blue	Orange / red	Red or orange
Purple / magenta	Green	Green

The practical rule — **use a filter of the colour the solution absorbs, which is the *complement* of the colour it appears** — is counter-intuitive to pupils and is the single most valuable piece of technique in the activity.

5.2 The broadband limitation, and how to work around it honestly

The sensor integrates across the visible band with a response resembling the eye's. It therefore gives one number in which all wavelengths are mixed together. Two consequences follow.

First, **absorbance measured with white light is not a true absorbance at any wavelength**: it is a weighted average, and the weighting depends on the spectrum of your lamp. Two groups using different lamps will get different numbers for the same solution. This is not a fault to be hidden — it is a real property of broadband photometry and should be measured deliberately (investigation 5).

Second, the fix: **put a coloured filter in the light path** so that only a band of wavelengths reaches the sensor. The instrument then becomes a *filter photometer*, and its readings do correspond to absorption within that band. Filter photometers were the standard analytical instrument for decades and are still used in field kits; a school colorimeter is exactly this. Choosing the filter that matches the solution's absorption band improves sensitivity by a large factor, and demonstrating that improvement is a genuinely satisfying experiment.

What this instrument can never do is scan wavelength and produce an absorption spectrum. If a pupil asks for the spectrum of chlorophyll, the correct answer is that this instrument cannot provide it, that a diffraction grating and a position-sensitive detector would be required, and that a set of filters gives a crude sampled version of the same information — a handful of points instead of a curve.

6. Ready-made investigations

6.1 Investigation 1: how much light does each filter let through?

Research question. How does transmittance vary between coloured filters, and what does that tell us about the light source?

Method. Build the fixed rig, lamp on, room light excluded, nothing in the light path. Press **Set $\mathbf{I_0}$ (reference)** and record the value. Then insert each filter in turn at the same fixed slot, wait five seconds and record the transmittance and the absorbance. **Between filters return to a clear path and check that the reading comes back to I_0** — if it does not, something has moved.

What to measure. Lux, transmittance and absorbance for each filter colour, plus the clear-path check after each.

Expected finding. Transmittance varies widely, typically from 60–80% for a pale yellow gel down to 5–15% for a deep blue. **Green and yellow filters transmit far more *measured* lux than blue or red ones** of comparable visual density, because the sensor is most sensitive in the green — an important and initially puzzling result that leads straight into the discussion of detector response. Two filters stacked give a transmittance close to the product of the individual values, and absorbances that add: an excellent check of the logarithm.

6.2 Investigation 2: the dilution series and the Beer–Lambert law

Research question. Is absorbance proportional to concentration?

Method. Prepare a stock solution of red or magenta food colouring, or of very dilute permanganate. Using syringes, make **from the same stock** at least five dilutions of known relative concentration — for example 100%, 80%, 60%, 40% and 20% — plus pure water as the blank. (A "1:2" dilution — one part solution, one part water — *halves* the concentration.) Fill identical containers to identical depths. **Set the reference with the water blank in the slot**, not with an empty slot: this takes the container and the solvent out of the measurement, which is what a real analyst does. Measure in increasing order of concentration, returning to the blank periodically to check for drift.

What to measure. Absorbance for each concentration. Plot A against relative concentration.

Expected finding. A good straight line through the origin over the lower part of the range, typically up to about $A = 1$ (with white light often sooner), then a downward curve, because the light is polychromatic and because stray light begins to dominate. **Pupils should identify the linear region themselves and state the range over which their instrument is trustworthy** — a professional habit worth building. Fit a line to the linear region and use it to determine the concentration of an unknown you prepared in advance.

6.3 Investigation 3: choosing the right filter (the colorimeter experiment)

Research question. Does a filter matched to the solution improve the measurement?

Method. Take two solutions of contrasting colour, for example a blue one and a yellow one, each at two concentrations that are visibly but not dramatically different. Measure the absorbance of each, first with white light and then through each of three filters — say red, green and blue — placed in the beam. **The reference must be reset for every filter**, with the blank in the slot and that filter fitted.

What to measure. For each solution-filter combination, the absorbance, and more importantly the *difference* in absorbance between the two concentrations of that solution.

Expected finding. The filter whose colour is complementary to the solution's appearance gives by far the largest difference between concentrations — often several times larger than white light. A blue solution measured through a red filter separates its concentrations cleanly; the same solution through a blue filter barely responds, because the blue light passes almost unabsorbed. **This is the working principle of every colorimeter, discovered by the pupils themselves.**

6.4 Investigation 4: path length, and the inverse-square check

Research question. Does the amount of solution the light travels through matter as much as the concentration?

Method. Part one: with a single solution, measure the absorbance through containers of different widths — or, more simply, by stacking one, two and three identical thin containers of the same solution in the beam. Part two, a useful physics control: with no sample, measure the lux at several measured distances from the lamp and plot intensity against $1/d^2$.

What to measure. Absorbance against the number of containers (that is, against path length); and lux against distance.

Expected finding. Absorbance rises linearly with path length, exactly as $A = \varepsilon c l$ predicts, which shows pupils that **doubling the concentration and doubling the thickness have the same effect**. The distance experiment gives a good straight line against $1/d^2$ provided room light has been excluded, and it vividly explains why the lamp must not move once the reference is set: **a** 10% **change in distance changes** I_0 **by about** 20%**.**

6.5 Investigation 5: reproducibility across days and lamps

Research question. Would we get the same answer next week, or with a different lamp?

Method. Repeat the full calibration of investigation 2 on three occasions: the same rig on a different day; the same day with a different lamp (LED versus filament, or torch versus desk lamp); and the same day with the room lights on rather than off. Nothing else may change. Export after each run; the date-stamped files make the comparison straightforward.

What to measure. The gradient of the A against c calibration line in each case, and the absorbance of one fixed reference solution measured in all conditions.

Expected finding. The *shape* of the relationship is robust — absorbance stays proportional to concentration — but the *gradient* changes with lamp type, because different lamps emit different spectra and the solution absorbs them unequally. Leaving the room lights on flattens the whole calibration, because stray light that never passed through the sample raises I and suppresses the apparent absorbance. Pupils conclude that **a broadband photometer measures a quantity that depends on its own light source**, and that calibration must be repeated whenever anything changes. This is the honest, grown-up finding of the whole activity.

The investigation, in one table

The question. Is absorbance proportional to concentration — and how far?
The hypothesis. Written **before** any measuring: if I dilute by half, the absorbance will _____, plus a reason. It is not changed afterwards.
What I change. One thing only: the **concentration** of the solution in the container.
What I measure. Absorbance, from the **Absorbance** card, after five seconds of settling — and I write I_0 down on paper, because it does not appear in the exported file.
What I keep the same. The same lamp at the same distance; the same container filled to the same depth; the same position in the slot; the dark box closed every time; and **the same reference**, set with the water blank in the slot.
How many repeats. One reading per concentration, plus **a return to the blank every two or three samples**. The blank must give an absorbance close to zero every time; if it stops doing so, the reference has gone stale and the series before it is no longer comparable with the series after.
What makes a reading valid — and what makes it an artefact. Valid is a reading taken with the box closed, the container wiped and held by the top, the reference fresh and the absorbance below 1. An artefact is anything else: room light reaching the sensor without passing through the sample flattens the whole scale towards zero; a torch on fading batteries makes every later absorbance read high; a bubble or a fingerprint changes the transmission by more than a dilution does. **And if the sample gives more light than the blank ($I > I_0$), the reference is no longer valid — reset it, do not report it.**

7. Analysis after the activity

7.1 What the data should look like

Sample	Transmittance	Absorbance	Comment
Clear path (reference)	100%	0.00	By definition.
Empty container	88–97%	0.01–0.06	Reflection at the surfaces — the reason to use a blank.
Container of water (blank)	85–95%	0.02–0.07	Set the reference here for work with solutions.
Pale filter (yellow, pale green)	50–80%	0.1–0.3	
Mid filter (orange, cyan)	20–50%	0.3–0.7	
Deep filter (blue, deep red)	3–15%	0.8–1.5	
Two filters stacked	product of the two	sum of the two	The logarithm check.
Dilute dye	40–80%	0.1–0.4	The linear region.
Concentrated dye	under 5%	above 1.3	Beyond the linear region; do not trust it.

Absolute values depend on the lamp, the geometry and the dyes used. **Compare trends, gradients and ratios; do not compare absolute absorbances between groups using different rigs.**

7.2 Questions to ask afterwards

1. Your solution looks blue. Which filter gave you the biggest change in absorbance when you diluted it, and why is it not the blue one?
2. Two filters stacked gave transmittances of 40% and 25% individually. Predict the transmittance and the absorbance of the pair, and explain why absorbances add while transmittances multiply.
3. Your calibration line was straight up to $A = 1$ and then bent over. Give two distinct explanations, one about the solution and one about the instrument.
4. Somebody opened the blinds halfway through your dilution series. Which way would your measured absorbances be biased, and why?
5. Our sensor cannot tell red light from green light — it reports a single number. Describe what would have to be added to the instrument to turn it into a true spectrometer, and state one question you could then answer that you cannot answer now.

7.3 Assessment ideas

- **Low stakes:** the pupil ranks five filters by transmittance and states, in one sentence, which colour of solution each filter would be best for measuring. **No logarithms** — so it is accessible in year IX too.
- **Standard:** a full calibration write-up — the dilution table, absorbance against concentration with a line of best fit, the linear range identified, and an unknown determined by interpolation with an estimate of its uncertainty.

- **Extension:** the multi-lamp reproducibility study, with a written evaluation of what a broadband detector can and cannot measure, a recomputation of T and A from the raw exported lux column, and a proposal for improving the rig towards a true spectrometer.

Common pitfalls and misconceptions

Forgetting to reset the reference. Every change of lamp, filter, distance or container type invalidates I_0. **If the absorbance of the blank is not close to zero, the reference is stale.**

Referencing against an empty slot instead of a blank. The container and the solvent both absorb and reflect a little. Set the reference with a container of pure water in place, and the measurement then reports the dye alone.

Ambient light. This ruins more runs than anything else. Room lights, sunlight through a window, a laptop screen and a passing cloud all reach the sensor without passing through the sample, which compresses the absorbance scale towards zero. Use the dark box every time.

A dimming torch. Batteries fade measurably over a twenty-minute experiment, so I_0 drifts downwards and every later absorbance reads high. Use a mains lamp, or re-check the blank every few samples.

Assuming the app produces a spectrum. It produces one number. Say so in the first minute.

Expecting the app's absorbance to match a textbook value. It is a broadband, lamp-dependent quantity, not a molar absorbance at a stated wavelength.

Milk, orange juice and other cloudy samples. They reduce the light by *scattering*, not absorption, and some of the scattered light still reaches the sensor, depending on how close it sits. They are not examples of Beer–Lambert; they are a good introduction to turbidimetry, if you say what the difference is.

Mismatched or scratched containers. Fingerprints, bubbles, scratches and different wall thicknesses all change the transmission. Handle containers by the top, wipe them, and use the same one for the whole series where possible.

Working outside the linear range. Very dark solutions give unstable, compressed readings. Dilute until the absorbance falls below about 1.

Sensor saturation. Before firmware 2.2.5 the reading stopped at about 3800 lx, and a nearby desk lamp easily exceeds that. **If I_0 sits at about 3800 and barely changes when you insert a pale filter, back the lamp off** or reflash the box.

Safety and classroom management

Dyes and solutions. Food colouring and ink are low hazard but stain skin, clothing and benches — gloves, aprons and immediate mopping of spills. **Potassium permanganate is an oxidiser and stains badly**: use only very dilute solutions prepared by the teacher, with eye protection, and never let pupils handle the solid. Copper sulfate solution is harmful if swallowed and an environmental hazard; follow local disposal rules and do not pour concentrated solutions down the drain. **Nothing in this activity is ever tasted**, including tea and squash used as samples.

Eye protection is worn whenever solutions are poured, diluted or transferred.

Lamps become hot, particularly filament and halogen types. Keep them clear of card, cloth and plastic, never cover them, and let them cool before packing away. Do not shine a bright lamp or torch into anyone's eyes, and **use no laser pointers** in place of the lamp.

Liquids and electronics. The box, the laptop and the mains lamp are kept clear of the pouring area. Fill containers away from the rig, wipe them dry, and then place them in the beam. A spill onto a powered lamp or laptop is the realistic worst case here. **The box is not waterproof**: no sample is held above it.

Glassware. Prefer plastic cuvettes and containers. If glass is used, check for chips and have a procedure ready for sweeping up.
Dark boxes reduce visibility on the bench. Keep the working area tidy and well lit outside the box itself.

Adaptations for pupils with special educational needs

Investigation 1 alone, with no logarithms. This is the main adaptation and it is not a concession: the pupil inserts each filter and records the *percentage* from the **Transmittance** card, then ranks the filters. Transmittance is a plain ratio, so the lesson is complete for anyone who does not yet have logarithms in their syllabus — that is, for the whole of year IX.
The rig built in advance and the reference set by the teacher. For pupils whose whole lesson would go on building the rig, give it ready-made with I_0 already set. The task that remains is measuring and interpreting, which are the skills being assessed.
The dilution table, pre-filled. The column of relative concentrations (100%, 80%, 60%, 40%, 20%) and the volumes of water and stock for each arrive printed on the sheet. The pupil measures with the syringe and records the absorbance; the arithmetic of dilution is no longer in the way.
Axes drawn in advance. For anyone whose whole lesson goes on constructing a graph, give a sheet with the axes drawn, scaled and labelled. The idea being learned is *linearity and where it is lost*, not the drawing of axes.
Three real roles for pupils with motor difficulties. Not being able to handle a syringe does not mean not taking part. The roles rotate at every sample:

- **the blank keeper** — decides when to return to the blank and has the right to stop the series if the blank's absorbance is no longer close to zero; this is the role with the greatest scientific value in the group, because they hold the validity of the whole calibration;
- **the screen reader** — says the absorbance aloud and announces immediately if the app asks for the reference to be set again;
- **the scribe** — writes the values down and plots the points as they arrive, which makes the bend in the curve visible during the lesson rather than at home.

Sensory sensitivity. The dark box, laboratory smells and the lamp's brightness can all be unpleasant. The scribe and screen reader roles are done at a distance, and the lamp can be shielded at the sides with card so that it lights only the inside of the box.

If things do not go to plan

You have no coloured filters. Coloured acetate, transparent coloured folder covers and even sweet wrappers work, flattened and taped over a card with a window cut in it. They are not calibrated — but they do not need to be: the activity compares filters with each other, it does not measure them absolutely.
You have no identical cuvettes. Use the same container for the whole series, rinsed between samples and wiped. That is actually better than "identical" containers that are not.
You have no dark box. Switch the lights off, close the blinds and lay a black cloth over the rig. If even that is impossible, **measure the background with the lamp off and subtract it** from every reading before computing T — it is less good, but it is honest, and pupils learn from it what stray light means.
The app asks you to set the reference again ($I > I_0$). It is not a fault. It means the sample is letting through more light than the reference — the absorbance would be negative, which is why the app leaves the cards blank: the lamp has warmed and brightened, something has moved, or room light has crept in. Reset the reference with the blank and repeat the

last samples.
You press "Set I_0" and nothing happens. The reading at that moment was zero: the lamp is off, the sensor is covered, or the box is not sending data. The line stays on "not set". Switch the lamp on, check the connection and try again.
The box will not connect. Do not stop the lesson. Use the "What the data should look like" table as a printed dataset: pupils compute A from T for each row, check the rule for two stacked filters (transmittances multiply, absorbances add) and plot the calibration line from the given values. **All the mathematics of the lesson is unchanged** — and it is the hard part. Measure again another day, after checking the cable, the port selected in the app and, if you work over WiFi, the network.

8. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **Royal Society of Chemistry** — `https://www.rsc.org` — search for colorimetry and the Beer–Lambert law in their schools material; they have classroom-tested versions of the dilution series with easily obtained substances.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the Beer's law simulation; it lets pupils change concentration, path length and wavelength and watch the absorbance respond, either just before or just after they measure it themselves.
- **Institute of Physics** — `https://www.iop.org` — search for material on light intensity and absorption; useful for the physics half of investigation 4.
- **Nuffield Foundation** — `https://www.nuffieldfoundation.org` — classic, carefully trialled practical work; useful for the low-tech versions that run alongside the measurement.
- **Royal Society of Biology** — `https://www.rsb.org.uk` — for the link to pigments and chlorophyll, if you go on to leaf extracts.
- **NIST** — `https://www.nist.gov` — search for photometry and spectrophotometry standards, if a pupil asks how an instrument of this kind is really calibrated.

9. Assessment

The exit ticket, identical to the one on the pupil sheet: *why does an absorbance mean nothing without a reference?* Collected in the last two minutes, it says more about the class's understanding than any mark.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Scientist level the three axes look like this:

- **Measurement practice** — the pupil sets the reference with the blank rather than an empty slot, returns to the blank periodically, keeps the box closed, and recognises the $I > I_0$ warning as a stale reference.
- **Data interpretation** — the pupil plots A against c, *identifies the linear region themselves* and states the range over which their instrument is trustworthy, instead of drawing one line through every point.
- **Scientific reasoning** — the pupil explains why the complementary filter gives the greatest sensitivity, and why an absorbance measured with white light depends on the lamp and cannot be compared with a textbook value.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

10. Pupil sheet

The sheet the pupils fill in is a separate document: `docs/en/student_sheets/spectroscopy.pdf`. It has two pages — the rig, the reference and the filter series on the first; the dilution series, the calibration curve and the conclusion on the second — so it is printed double-sided, one sheet per pupil. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write.

Three things to know before printing it. **The sheet asks for I_0 to be written by hand, in a box of its own**, and this is not a formality: the reference value *does not appear in the exported file*, and without it nobody can recompute the transmittance and the absorbance at home. **The dilution table arrives with the volumes already printed** for each relative concentration, so that the arithmetic of dilution does not eat the measuring time. **The table has a blank row at the start and a second, control, blank in the middle of the series**, printed as such, because returning to the blank is what makes the series valid — and if the control no longer gives close to zero, the pupil can see on the sheet exactly between which samples the series went wrong. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.